

Exercise 1. Solve exercise 2 on page 230.

Exercise 2. Let $T : C([0, 1]) \rightarrow C([0, 1])$ be defined by

$$Tf(x) = \frac{1}{x} \int_0^x f(t)dt, \quad x > 0 \quad Tf(0) = f(0).$$

Show that T is bounded and determine its spectrum.

Hint: The equation $\lambda f - Tf = g$ leads to a relatively simple differential equation for $F(x) = \int_0^x f(t)dt$.

Exercise 3. Let $u : \mathbb{R} \mapsto \mathbb{C}$ be a bounded continuous function and consider the bounded operator M_u of multiplication by u on $L^2(\mathbb{R})$,

$$M_u f = uf.$$

Show that $\sigma(M_u) = \overline{u(\mathbb{R})}$.